

☒ Core and Distributed Network Solutions – Detailed Theory

☒ 1. Core Network Solutions

Core network solutions are central to any enterprise network and are designed to provide high-speed, reliable, and secure connectivity between various components like servers, data centers, storage systems, and gateways.

☒ Key Features of Core Networks:

- High throughput
- Low latency
- Centralized traffic management
- Redundant design
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Scalability for future growth

☒ Technologies in Core Networks:

- Routing and Switching: High-end routers and Layer 3 switches.
- VLANs & Trunking: Efficient network segmentation and traffic handling.
- Dynamic Routing Protocols: OSPF, EIGRP, BGP.
- Load Balancing & Failover: Using devices like F5 or HA-pairs.
- QoS (Quality of Service): Prioritization of traffic like VoIP or video.

☒ 2. Distributed Network Solutions

Distributed networks decentralize computing resources by placing services close to users (e.g., in branch offices), improving performance, availability, and fault tolerance.

☒ Why Use Distributed Networks?

- Reduces latency for remote offices
- Enhances fault tolerance
- Local data processing and storage
- Better use of WAN links

☒ Common Use Cases:

- Branch office deployments

- Multi-site enterprise setups
 - Global content distribution
 - Hybrid cloud integration
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☒ 3. IPv4 and IPv6 Addressing

☒ IPv4:

- 32-bit address space (e.g., 192.168.1.1)
- Supports ~4.3 billion unique addresses
- Written in dotted decimal notation

- Subnetting divides networks for efficiency

☒ IPv6:

- 128-bit address space (e.g., 2001:0db8:85a3::8a2e:0370:7334)
- Vastly larger address space ($\sim 3.4 \times 10^{38}$)
- Hexadecimal and colon-separated
- Integrated IPsec, no need for NAT

Feature	IPv4	IPv6
Address Length	32 bits	128 bits
Notation	Decimal (e.g., 192.0.2.1)	Hexadecimal (e.g., fe80::1)

NAT Required	Yes	No
Security	Optional (IPsec)	Mandatory (IPsec supported)

❏ 4. Distributed File System (DFS)

DFS allows administrators to group shared folders located on different servers and present them to users as a single namespace.

❏ Types of DFS:

- DFS Namespaces: Logical view of shared folders in a unified path (\\domain\dfsroot)
- DFS Replication (DFSR): Keeps files in sync across multiple locations using multi-master replication

❏ Use Case:

- Provide consistent file access in headquarters and branch offices
 - Enable fault-tolerant file access
 - Improve performance for local file access
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☒ 5. Branch Office Solutions

A branch office setup includes extending IT services such as file access, print services, and secure internet to remote or smaller offices connected via WAN.

☒ Key Technologies:

- Read-Only Domain Controller (RODC): Domain controller with read-only Active Directory data
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DFS Replication: For local file access

- **BranchCache:** Caches content from central servers
- **DHCP and DNS:** May be locally deployed or forwarded to HQ
- **VPN:** Secure site-to-site or client VPN for branch connectivity

☒ **Benefits:**

- Enhanced performance and autonomy for branches
 - Secure and managed access to HQ resources
 - Minimal need for full IT staff at branch sites
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☒ Lab Activities Overview

Lab 1: Implement IPv4 and IPv6 Addressing

1.
Assign static IPs to interfaces (IPv4 and IPv6)
 2.
Verify routing with ping, tracert, and ipconfig
 3.
Setup and test dual-stack (IPv4 + IPv6)
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Lab 2: Implement DFS and Replication

1.
Install DFS Namespace and DFSR roles via Server Manager

2.
Create a DFS Namespace (e.g., \\corp.local\dfs)
 3.
Add targets (e.g., shared folders on other servers)
 4.
Configure replication groups and schedules
 5.
Validate replication using DFS Management and event logs
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Lab 3: Deploy Branch Office Solution

1.
Setup a Branch Office server (RODC or member server)
2.
Join it to the domain and install DFS Replication and BranchCache
3.
Configure DHCP relay or deploy DHCP locally

4.

Setup DNS forwarding if DNS is not local

5.

Simulate a WAN outage to test offline availability